Mass Gap for the O(N) Model: Analysis of the HS Approach

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Abstract

We analyze the Hubbard-Stratonovich approach to proving the mass gap for the O(N) model in two dimensions at large N. The HS identity for the Euclidean quartic requires imaginary coupling, creating a sign problem. We compare the NLSM (where Kupiainen's constraint trick gives a real mass m_0^2 in the operator, enabling a direct FK bound for each σ-value) with the LSM (where the real mass cancels after HS, and the mass gap must emerge from the oscillatory σ-average). A comparison across $d = 0, 1, 2$ shows that the sign problem is specific to $d = 2$: the logarithmic UV divergence of the Green's function promotes the self-energy from $O(1/N)$ to $O(1)$, causing dimensional transmutation and making the oscillatory cancellations essential. We discuss three proof strategies and the current status of the Lean formalization.

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1 The HS identity and the sign problem

1.1 The Euclidean quartic requires imaginary coupling

The O(N) LSM action on $\Lambda \subset \mathbb{Z}^2$:

$$S[\varphi] = \tfrac{1}{2}\sum_i \langle \varphi^i, (-\Delta)\varphi^i\rangle + N\sum_x \lambda\big(\tfrac{|\varphi(x)|^2}{N} - v^2\big)^2. \tag{1}$$

The Boltzmann weight is e^{-S}. The quartic appears as $e^{-\lambda a^2}$ with $a = |\varphi|^2/N - v^2$. The HS identity:

$$e^{-\lambda a^2} = \frac{1}{\sqrt{4\pi\lambda}}\int_{-\infty}^{\infty} e^{-\sigma^2/(4\lambda)\,+\,i\sigma a}\,d\sigma. \tag{2}$$

Here $\sigma \in \mathbb{R}$ (real integration variable), $e^{-\sigma^2/(4\lambda)}$ ensures convergence, and the coupling $i\sigma a$ is **imaginary**. This is a Fourier transform (proved from Mathlib's `fourierIntegral_gaussian`).

The imaginary coupling is forced by the *minus* sign in $e^{-\lambda a^2}$ (repulsive/confining quartic in Euclidean signature). A real coupling would give $e^{+\lambda a^2}$ (attractive, wrong sign).

1.2 The σ-integral on the real axis

After applying (2) at each site and integrating out φ (Gaussian with complex mass), the partition function becomes:

$$Z = c\int_{\mathbb{R}^{|\Lambda|}} \det\big(-\Delta + 2i\sigma z\big)^{-N/2}\, e^{-\sum_x[\sigma(x)^2/(4\lambda)+i\sigma(x)v^2]}\, d^{|\Lambda|}\sigma, \tag{3}$$

where $z = N^{-1/2}$ and the det comes from the φ-integral.

This integral **converges** on the real σ-axis:

- $e^{-\sigma^2/(4\lambda)}$ provides Gaussian decay in σ.
- $|\det(-\Delta + 2i\sigma z)^{-N/2}| \leq \det(-\Delta)^{-N/2}$ (bounded, since $|1 + i\theta| \geq 1$ for real θ).

But the integrand is **complex** (the det and the $e^{i\sigma v^2}$ factor oscillate). This is the sign problem.

2 The NLSM: Kupiainen's mass trick

2.1 Adding m_0^2 for free

For the NLSM with constraint $|\varphi|^2 = N\beta$: any function of $|\varphi|^2$ is constant on the constraint surface. In particular, $\frac{1}{2}m_0^2|\varphi|^2 = \frac{1}{2}m_0^2 N\beta = \text{const}$.

So we can **freely add** $\frac{1}{2}m_0^2|\varphi|^2$ to the action, giving the massive GFF:

$$S_{\text{NLSM}} = \tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle + \text{const}, \qquad |\varphi|^2 = N\beta. \tag{4}$$

The parameter $m_0^2 > 0$ is arbitrary; we choose it via the gap equation (Kupiainen eq. 9b): $(-\Delta + m_0^2)^{-1}_{ii} = \beta$.

2.2 The NLSM operator has real mass

After HS (introducing the Lagrange multiplier α for the constraint), the φ-propagator at fixed α is:

$$\big(-\Delta + m_0^2 - 2i\alpha z\big)^{-1}(x, 0). \tag{5}$$

The operator has **real part** $-\Delta + m_0^2 \geq m_0^2 > 0$. By the triangle inequality (FK representation):

$$\big|(-\Delta + m_0^2 - 2i\alpha z)^{-1}(x, 0)\big| \leq (-\Delta + m_0^2)^{-1}(x, 0) \sim e^{-m_0|x|}. \tag{6}$$

Every α-value gives exponential decay with mass m_0. The α-average is a correction, not the leading effect.

2.3 Kupiainen's proof structure

1. Gap equation: $m_0 > 0$ (always in 2d).
2. Bound (6): each α-value gives decay.
3. Cluster expansion + RP: control the α-average (prefactor, not the mass).
4. Mass gap: $m = m_0(1 + O(1/N))$.

The hard step is (3), but the mass m_0 comes for free from (2).

3 The LSM: the mass cancels

3.1 Add-subtract fails

For the LSM (no constraint), we try the same trick: add $\frac{1}{2}m_0^2|\varphi|^2$ to the action and subtract it from the interaction:

$$S = \left[\tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle\right] + \left[\lambda(u - v^2)^2 - \tfrac{1}{2}m_0^2 u\right], \qquad u = |\varphi|^2/N. \tag{7}$$

Apply HS to $\lambda(u - v^2)^2$ only. The subtracted mass $-\frac{1}{2}m_0^2 u$ remains outside the HS. Computing the φ-dependence:

$$\begin{aligned} &-\tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle + i\sigma|\varphi|^2 + \tfrac{1}{2}m_0^2|\varphi|^2 \\ &\quad = -\tfrac{1}{2}\langle\varphi, (-\Delta)\varphi\rangle + i\sigma|\varphi|^2 \\ &\quad = -\tfrac{1}{2}\langle\varphi, (-\Delta - 2i\sigma)\varphi\rangle. \end{aligned} \tag{8}$$

The m_0^2 from the GFF and the $-m_0^2$ from the subtraction **cancel exactly**. The LSM operator is:

$$-\Delta - 2i\sigma \qquad (\textbf{no real mass}). \tag{9}$$

3.2 Consequences

The operator (9) has eigenvalues $k^2 - 2i\sigma$ with real part $k^2 \geq 0$ (from $-\Delta$) and imaginary part -2σ. For $k = 0$: the eigenvalue is $-2i\sigma$ (purely imaginary).

- **No single-σ decay**: $|(-\Delta - 2i\sigma)^{-1}(x, 0)|$ does NOT decay exponentially for each σ-value. At $k = 0$: $|1/(-2i\sigma)| = 1/(2|\sigma|)$, independent of $|x|$.
- **Mass from σ-average only**: The exponential decay of $\langle\varphi(x)\varphi(0)\rangle$ must come *entirely* from the oscillatory cancellations in the σ-integral (3).
- **The FK bound fails**: The NLSM bound (6) relies on the real mass m_0^2. Without it, $|G_\sigma| \leq G_0$ gives $G_0 = (-\Delta)^{-1}$ (massless propagator, no decay).

3.3 Comparison

	NLSM	LSM
Mass in operator	m_0^2 (free, from constraint)	**none** (cancels)
Operator after HS	$-\Delta + m_0^2 - 2i\alpha z$	$-\Delta - 2i\sigma$
Single-σ decay	$e^{-m_0\|x\|}$ (yes)	**none**
Mass gap source	Real mass + corrections	**σ-average only**
FK bound	Works ($\|G\| \leq G_{m_0}$)	**Fails**

4 Comparison across dimensions

The sign problem and mass gap structure depend strongly on dimension.

4.1 $d = 0$ (matrix model)

One site ($|\Lambda| = 1$). The σ-integral is one-dimensional: $\int_{-\infty}^{\infty} e^{-\sigma^2/(4\lambda)+i\sigma a}\, d\sigma$. The saddle-point Gaussian is exact up to $O(1/N)$ corrections. Oscillations are $O(1/N^2)$ — negligible. No sign problem.

The Green's function $C = 1/m_0^2$ is a number. The self-energy $\Sigma = O(1/N)$. The mass is perturbative: $m = m_{\text{bare}}(1 + O(1/N))$. No dimensional transmutation.

4.2 $d = 1$ (quantum mechanics)

$|\Lambda| = L/a$ sites (discretized time). The Green's function $C(t, t) = 1/(2m_0)$ is **finite** (no UV divergence). The self-energy $\Sigma = O(1/N)$ (no log divergence).

Oscillations at the saddle width: $\sim L/(aN^2)$. For fixed L, a and large N: oscillations vanish. The saddle-point expansion converges for $N > C \cdot L/a$. The mass gap is perturbative.

4.3 $d = 2$ (QFT)

$|\Lambda| = (L/a)^2$ sites. The Green's function $C(x, x) = \frac{1}{4\pi} \log \frac{\Lambda^2}{m_0^2}$ **diverges** logarithmically with the UV cutoff.

The self-energy: $\Sigma \sim \frac{1}{N} \cdot \log \frac{\Lambda}{m_0} \sim \frac{1}{N} \cdot N = O(1)$. The $1/N$ from BL is compensated by the log divergence — **dimensional transmutation**. The gap equation gives $m_0 \sim \Lambda\, e^{-c/g^2}$ (exponentially small).

Oscillations at the saddle width: $\sim |\Lambda|/N^2 = L^2/(a^2N^2)$. For the continuum limit $a \to 0$: oscillations grow, sign problem becomes severe.

4.4 Summary

	$d = 0$	$d = 1$	$d = 2$
$C(x, x)$	$1/m_0^2$	$1/(2m_0)$	$\frac{1}{4\pi} \log \frac{\Lambda}{m_0}$
UV divergence	none	none	**log**
Self-energy Σ	$O(1/N)$	$O(1/N)$	$\mathbf{O(1)}$
Gap equation	trivial	perturbative	**dim. transmutation**
Oscillation at width	$O(1/N^2)$	$O(L/(aN^2))$	$O(L^2/(a^2N^2))$
Sign problem	none	mild	**severe**

The fundamental difference: in $d = 2$, the log divergence of $C(x, x)$ makes the self-energy $O(1)$ instead of $O(1/N)$, producing dimensional transmutation and making the sign problem unavoidable.

5 The saddle-point structure

Despite the difference in operator structure, both models share the same saddle-point analysis.

5.1 The gap equation

On the real σ-axis, the saddle of the exponent in (3) is at $\sigma = 0$ (after absorbing the saddle value into m_0^2 for the NLSM, or into the gap equation for the LSM). The gap equation in both cases:

$$(-\Delta + m_0^2)_{xx}^{-1} = \beta_{\text{eff}} \tag{10}$$

where β_{eff} encodes the coupling. In 2d: $m_0 \sim \Lambda\, e^{-c/g^2}$ (exponentially small).

5.2 The saddle-point Gaussian

Near $\sigma = 0$: the exponent is approximately $-\frac{1}{2}(\sigma, H\sigma)$ where $H > 0$ (positive definite Hessian, including both the bare $1/(2\lambda)$ and the $\operatorname{Tr}\log$ contribution). The saddle-point Gaussian IS log-concave on the real axis.

5.3 The oscillatory corrections

Away from $\sigma = 0$: the exponent develops an imaginary part $\operatorname{Im} f(\sigma) \neq 0$, causing oscillations. The sign problem is the difficulty of controlling these oscillations.

Key scales (at the saddle):

- Width of the Gaussian: $\sigma \sim 1/\sqrt{N}$ per site.
- Oscillation onset: $\operatorname{Im} f \sim O(\sigma^3/\sqrt{N})$ (cubic, since $\operatorname{Im} f'' = 0$ at the saddle).
- Total oscillation at the saddle width: $\sim |\Lambda|/N^2$.

For $N \gg \sqrt{|\Lambda|}$: oscillations are negligible within the saddle-point width. The saddle-point approximation is accurate.

6 What Kupiainen proved

6.1 NLSM (1980a): mass gap

Kupiainen [2] proved the mass gap for the lattice NLSM at large N: exponential decay of correlations in the odd sector, with mass $m(\beta, n) \to m_0(\beta)$ as $n \to \infty$.

The proof uses the constraint trick (m_0^2 added for free), the dual representation with complex measure, reflection positivity, chessboard estimates, and a random walk expansion. The threshold is $n > a_1 m_0^{-a_2}$ with $a_2 = 6(d+2) + 1 = 25$ in $d = 2$.

6.2 LSM (1980b): pressure only

Kupiainen [3] studied the continuum $(\varphi^2)_2^2$ model (the LSM) and proved a **weaker** result: the $1/n$ expansion for the **pressure** (vacuum energy) is asymptotic:

$$p = \sum_{j=0}^{m-1} p_j n^{-j} + R_m, \qquad |R_m| \leq (3m)!\, \lambda^4 \alpha^m \Big(\frac{\lambda^\beta}{n}\Big)^m. \tag{11}$$

The effective small parameter is λ^β/n (with β close to 10), not $e^{c\lambda}/n$.

He did NOT prove the mass gap for the LSM. The paper works in the φ-representation (integration by parts, no HS) and in the dual representation (complex nonlocal measure, same sign problem). He mentions that the mass gap would need "a new cluster expansion which explicitly cancels the diverging pressure factors" — left as future work that was apparently never published.

6.3 Implications

- The **NLSM mass gap** at large N is proved (Kupiainen 1980a).
- The **LSM mass gap** at large N is **open** — only the pressure expansion is known (Kupiainen 1980b).
- Proving the LSM mass gap would be a **genuinely new result**, not a reformalization of existing work.
- The LSM and NLSM share the same IR physics, so the mass gap "should" transfer. But making this rigorous is nontrivial.

7 Proof strategies

7.1 Strategy A: NLSM (Kupiainen)

Work with the NLSM directly. The constraint gives real mass m_0^2 for free. The FK bound gives $|G| \le G_{m_0}$ (single-σ decay). Cluster expansion + RP control the σ-average.

Advantage: Real mass in the operator, clean FK bound.
Disadvantage: Cluster expansion needed (technically heavy).
Threshold: $N > C\, m_0^{-a_2}$ with $a_2 = O(d^2)$ (Kupiainen Thm. 1).

7.2 Strategy B: LSM saddle-point

Work with the LSM. No real mass in the operator, but the saddle-point Gaussian on the real σ-axis IS log-concave.

Idea: Apply BL to the saddle-point Gaussian (log-concave, real, positive) and control the oscillatory corrections as a perturbation.

Challenge: The mass gap comes entirely from the σ-average (no single-σ decay). Need to show the oscillatory corrections don't destroy the saddle-point mass.

Advantage: Avoids cluster expansion (BL for the Gaussian).
Disadvantage: No FK bound; need new estimates for the oscillatory σ-average.

7.3 Strategy C: LSM $\to$ NLSM equivalence

Prove that the LSM and NLSM have the same IR physics (they differ only at the σ-mass scale $M_\sigma \sim \sqrt{\lambda}\, v$), then use Strategy A.

Advantage: Gets the NLSM mass gap (with the real mass trick) from the LSM formulation.
Disadvantage: Proving IR equivalence rigorously is itself a major result.

8 The $N = 2$ obstruction

For $N = 2$: the BKT transition occurs at finite λ (Dario-Garban [4]), driven by vortices ($\pi_1(S^1) = \mathbb{Z}$). Any proof must exclude $N = 2$. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no BKT.

9 Current status

9.1 What is proved (in Lean)

- HS identity (2): from Mathlib's `fourierIntegral_gaussian`.
- Convergence on the real axis: $|e^{i\sigma a}| = 1$, $|e^{-\sigma^2/(4\lambda)}| \leq 1$.
- HS equivalence (one-site): $\int d\sigma\, e^{-S_{\text{HS}}} = c \cdot e^{-S_{\text{original}}}$ (1 sorry for complex arithmetic).
- Contour rotation algebra: $(i\sigma')^2 = -\sigma'^2$, $i \cdot a \cdot (i\sigma') = -a\sigma'$, etc.
- Continuum limit: OS0–OS2 for the O(N) LSM on T_L^2.

9.2 What is axiomatized

- Contour shift (Cauchy's theorem for rectangles): provable from Mathlib + PNT project.
- Matrix calculus (det C^∞, Jacobi's formula).
- FK sub-Gaussian bound (placeholder for the mass gap step).
- Continuum limit technicalities (4 axioms, porting from pphi2/gaussian-field).

9.3 Open questions

1. Can Strategy B work? Is BL on the saddle-point Gaussian sufficient to control the oscillatory corrections?
2. What is the threshold N_0 for each strategy?
3. Can the N=1 test case (P(φ)$_2$) validate Strategy B?

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